Online Book Store Analysis by (SQL)

Introduction

This project focuses on analyzing an **Online Book Store dataset** using **SQL** to extract meaningful business insights. The main objective of this analysis is to understand customer purchasing behavior, identify best-selling books, track revenue performance, and evaluate category-wise sales trends.

Using SQL queries, I performed data exploration and generated reports such as **total sales**, **top customers**, **most popular book categories**, **monthly revenue trends**, and **order frequency analysis**. This helps in improving business decision-making, managing inventory efficiently, and increasing sales through better marketing strategies.

Overall, this project demonstrates my ability to work with relational databases, write optimized SQL queries, and convert raw data into valuable insights for business growth.

1. **Retrieve all books in the "Fiction" genre**

select * from book where genre='Fiction'

order by price desc limit 5;

	book_id [PK] integer	title character varying	author character varying	genre character varying	published_year integer	price numeric (10,2)	stock integer
1	100	Synchronized client-server service-desk	James Alvarado	Fiction	1906	49.89	29
2	260	Business-focused methodical applicati...	Brian King	Fiction	1907	49.59	10
3	142	Multi-tiered responsive parallelism	Amanda Wilson	Fiction	1940	48.96	11
4	274	Automated systemic functionalities	Matthew Thomas	Fiction	1903	48.80	72
5	278	Exclusive asymmetric installation	Benjamin Flores	Fiction	1901	48.61	69

Find books published after the year 1950:

select * from book where published_year>1950 limit 5;

	book_id [PK] integer	title character varying	author character varying	genre character varying	published_year integer	price numeric (10,2)	stock integer
1	2	Persevering reciprocal knowledge u...	Mario Moore	Fantasy	1971	35.80	19
2	4	Customizable 24hour product	Christopher Andre...	Fiction	2020	43.52	8
3	5	Adaptive 5thgeneration encoding	Juan Miller	Fantasy	1956	10.95	16
4	6	Advanced encompassing implemen...	Bryan Morgan	Biography	1985	6.56	2
5	8	Persistent local encoding	Troy Cox	Science Fiction	2019	48.99	84

1. **List all customers from the Canada:**

select * from customer where country='Canada';

	customer_id [PK] integer	name character varying	email text	phone numeric	city character varying	country character varying
1	38	Nicholas Harris	christine93@perkins.com	1234567928	Davistown	Canada
2	415	James Ramirez	robert54@hall.com	1234568305	Maxwelltown	Canada
3	468	David Hart	stokesrebecca@gmail.c...	1234568358	Thompsonfurt	Canada

1. **Show orders placed in November 2023:**

select * from orders where order_date between '2023-11-01' and '2023-11-30'

order by total_amount desc limit 5;

	order_id integer	customer_id integer	book_id integer	order_date date	quantity integer	total_amount numeric (10,2)
1	245	386	97	2023-11-01	9	411.66
2	137	474	471	2023-11-25	8	363.04
3	19	496	60	2023-11-17	9	316.26
4	4	433	343	2023-11-25	7	301.21
5	213	325	447	2023-11-17	7	253.75

1. **Retrieve the total stock of books available:**

select sum(stock) from book;

	sum bigint
1	25056

1. **Find the details of the most expensive book:**

select * from book order by price desc limit 1;

	book_id [PK] integer	title character varying	author character varying	genre character varying	published_year integer	price numeric (10,2)	stock integer
1	340	Proactive system-worthy orchestrati...	Robert Scott	Mystery	1907	49.98	88

1. **Show all customers who ordered more than 2 quantity of a book:**

select * from orders where quantity>2;

	order_id integer	customer_id integer	book_id integer	order_date date	quantity integer	total_amount numeric (10,2)
1	1	84	169	2023-05-26	8	188.56
2	2	137	301	2023-01-23	10	216.60
3	3	216	261	2024-05-27	6	85.50
4	4	433	343	2023-11-25	7	301.21
5	5	14	431	2023-07-26	7	136.36

1. **Retrieve all orders where the total amount exceeds $20:**

select * from orders where total_amount>20;

	order_id integer	customer_id integer	book_id integer	order_date date	quantity integer	total_amount numeric (10,2)
1	1	84	169	2023-05-26	8	188.56
2	2	137	301	2023-01-23	10	216.60
3	3	216	261	2024-05-27	6	85.50
4	4	433	343	2023-11-25	7	301.21
5	5	14	431	2023-07-26	7	136.36

9.List all genres available in the Books table:

select distinct genre from book;

	genre character varying
1	Romance
2	Biography
3	Mystery
4	Fantasy
5	Fiction

10.Find the book with the lowest stock:

select * from book order by stock;

	book_id [PK] integer	title character varying	author character varying	genre character varying	published_year integer	price numeric (10,2)	stock integer
1	44	Networked systemic implementation	Ryan Frank	Science Fiction	1965	13.55	0
2	163	Object-based eco-centric challenge	Douglas Mccarthy	Non-Fiction	1905	19.11	0
3	127	Business-focused real-time benchmark	David Nelson	Science Fiction	1997	11.66	0
4	60	Robust eco-centric capacity	Brian Haney	Biography	1990	35.14	0
5	378	Future-proofed heuristic function	Samantha Mcclain	Romance	1903	6.01	0

11. Calculate the total revenue generated from all orders:

select sum(total_amount) as Revenue from orders

	revenue numeric
1	75628.66

12. Retrieve the total number of books sold for each genre:

select book.genre,sum(orders.quantity) as Total_Sold from book

join orders on

orders.book_id=book.book_id

group by 1 order by Total_Sold;

	genre character varying	total_sold bigint
1	Fiction	225
2	Biography	285
3	Non-Fiction	351
4	Romance	439
5	Fantasy	446

13. Find the average price of books in the "Fantasy" genre:

select round(avg(price),2) as Avg_price from book

where genre='Fantasy'**;**

	avg_price numeric
1	25.98

14. List customers who have placed at least 2 orders:

select customer.customer_id,customer.name,count(orders.order_id) as Orders from customer

join orders on

customer.customer_id=orders.customer_id

group by 1,2 having count(orders.order_id)>2;

	customer_id [PK] integer	name character varying	orders bigint
1	272	Carl Smith	3
2	22	Stacey Adams	3
3	107	Amy Hunt	4
4	166	John Delacruz	3
5	47	Debra Park	3

15. Find the most frequently ordered book:

select book.title,orders.book_id,count(orders.order_id) as Frequently_order from book

join orders on

orders.book_id=book.book_id

group by 1,2 order by Frequently_order desc limit 1;

	title character varying	book_id integer	frequently_order bigint
1	Pre-emptive intangible adapter	491	4

16. Show the top 3 most expensive books of 'Fantasy' Genre :

select * from book where genre='Fantasy'

order by price desc limit 3;

	book_id [PK] integer	title character varying	author character varying	genre character varying	published_year integer	price numeric (10,2)	stock integer
1	240	Stand-alone content-based hub	Lisa Ellis	Fantasy	1957	49.90	41
2	462	Innovative 3rdgeneration database	Allison Contreras	Fantasy	1988	49.23	62
3	238	Optimized even-keeled analyzer	Sherri Griffith	Fantasy	1975	48.97	72

17. Retrieve the total quantity of books sold by each author:

select book.author,sum(orders.quantity) as Total_Sold from book

join orders on

orders.book_id=book.book_id

group by 1 order by Total_sold desc;

	author character varying	total_sold bigint
1	Patrick Contreras	28
2	Melissa Taylor	27
3	Emily James	24
4	Thomas Trujillo	24
5	Ellen Doyle	23

18. List the cities where customers who spent over $30 are located:

select distinct customer.city,orders.total_amount from customer

join orders on

customer.customer_id=orders.customer_id

where orders.total_amount>30 order by total_amount desc;

	city character varying	name character varying	customer_id integer	total_amount numeric (10,2)
1	Smithborough	Christine Maldonado	17	491.50
2	Fieldsland	Jason Villegas	77	489.60
3	Thomaschester	Kevin Collins	226	489.60
4	Mendezburgh	Catherine Wilkins	397	486.70
5	Melissaside	Annette Garcia	284	480.30

19. Find the customer who spent the most on orders:

select customer.customer_id,customer.name,sum(orders.total_amount) Amount_Spend from customer

join orders on

customer.customer_id=orders.customer_id

group by 1,2 order by Amount_Spend desc limit 1;

	customer_id [PK] integer	name character varying	amount_spend numeric
1	457	Kim Turner	1398.90

20. Calculate the stock remaining after fulfilling all orders:

select book.book_id,book.title,book.stock,coalesce(sum(orders.quantity),0) as Order_Quantity,

book.stock-coalesce(sum(orders.quantity),0) as Remaning_Stock from book

left join orders on

book.book_id=orders.book_id

group by 1 order by book.book_id;

	book_id [PK] integer	title character varying	stock integer	order_quantity bigint	remaning_stock bigint
1	1	Configurable modular throughput	100	3	97
2	2	Persevering reciprocal knowledg...	19	0	19
3	3	Streamlined coherent initiative	27	5	22
4	4	Customizable 24hour product	8	0	8
5	5	Adaptive 5thgeneration encoding	16	8	8

Thank you